



11027 - The last gasps of cosmic titans

Cycle: 5, Proposal Category: GO

INVESTIGATORS

<i>Name</i>	<i>Institution</i>
Dr. Steve Schulze (PI)	Weizmann Institute of Science
Dr. Lin Yan (CoI) (CoPI) (US Admin CoI)	California Institute of Technology
Dr. Joel Pearson Johansson (CoI) (ESA Member) (CoPI)	Stockholm University
Prof. Avishay Gal-Yam (CoI)	Weizmann Institute of Science
Dr. Ofer Yaron (CoI)	Weizmann Institute of Science
Dr. Jesper Sollerman (CoI) (ESA Member)	Stockholm University
Prof. Claes Fransson (CoI) (ESA Member)	Stockholm University
Dr. Adam Miller (CoI)	Northwestern University
Anamaria Gkini (CoI) (ESA Member)	Stockholm University
Dr. Anders Jerkstrand (CoI) (ESA Member)	Stockholm University
Dr. Ragnhild Lunnan (CoI) (ESA Member)	Stockholm University
Dr. Conor Omand (CoI) (ESA Member)	Stockholm University
Dr. Nikhil Sarin (CoI) (ESA Member)	University of Cambridge
Dr. Ting-Wan Chen (CoI)	National Central University
Dr. Lindsey Kwok (CoI)	Northwestern University
Dr. Daniel Perley (CoI) (ESA Member)	Liverpool John Moores University
Jacob Wise (CoI) (ESA Member)	Liverpool John Moores University
Prof. Keiichi Maeda (CoI)	Kyoto University
Ms. Aiswarya Sankar.K (CoI)	National Central University
Dr. Avinash Singh (CoI) (ESA Member)	Stockholm University
Dr. Mansi Kasliwal (CoI)	California Institute of Technology

OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Observation Folder				
	1	SN2018ibb - MIRI - MRS - off-scene	MIRI Medium Resolution Spectroscopy	(5) SN2018ibb-back
	2	SN2018ibb - MIRI - MRS - on-scene	MIRI Medium Resolution Spectroscopy	(1) SN2018ibb
	3	SN2018ibb - MIRI - Imaging	MIRI Imaging	(6) SN2018ibb-ima
	4	SN2020abjc - MIRI - MRS - off-scene	MIRI Medium Resolution Spectroscopy	(7) SN2020abjc-back
	5	SN2020abjc - MIRI - MRS - on-scene	MIRI Medium Resolution Spectroscopy	(2) SN2020abjc
	6	SN2020abjc - MIRI - Imaging	MIRI Imaging	(8) SN2020abjc-ima
	7	SN2021yfq - MIRI - MRS - off-scene	MIRI Medium Resolution Spectroscopy	(11) SN2021yfq-back
	8	SN2021yfq - MIRI - MRS - on-scene	MIRI Medium Resolution Spectroscopy	(3) SN2021yfq
	9	SN2021yfq - MIRI - Imaging	MIRI Imaging	(12) SN2021yfq-ima
	10	SN2023iop - MIRI - MRS - off-scene	MIRI Medium Resolution Spectroscopy	(9) SN2023iop-back
	11	SN2023iop - MIRI - MRS - on-scene	MIRI Medium Resolution Spectroscopy	(4) SN2023iop
	12	SN2023iop - MIRI - Imaging	MIRI Imaging	(10) SN2023iop-ima

ABSTRACT

The nature of the Universe's most extreme stellar explosions remains unresolved. Some superluminous supernovae are excellent candidates for pair-instability supernovae (PISNe) and pulsational PISNe (PPISNe) - the predicted deaths of stars with >95 Msun that likely dominated early cosmic evolution. However, no definitive (P)PISN has been confirmed due to observational degeneracies: optical signatures are mimicked by magnetar-powered explosions. We propose to break this degeneracy through a comprehensive MIRI campaign targeting two orthogonal diagnostics. (1) Dust mass: due to their enormous ejecta masses, (P)PISN models predict prolific dust formation (1-30 Msun), orders of magnitude more than core-collapse SNe or magnetar-powered events. (2) [Ne II] 12.8m emission: Magnetar models uniquely predict strong late-time [Ne II] from continued energy

injection, while (P)PISNe predict no such emission. We propose deep imaging in eight MIRI filters plus medium-resolution spectroscopy for four compelling (P)PISN candidates at 2-8 years post-explosion. A detection of >1 Msun dust without [Ne II] would provide the first definitive (P)PISN confirmation. [Ne II] detection would prove magnetar-powered origins, demonstrating that (P)PISN-like transients can arise from alternative mechanisms. A non-detection of both signatures would be equally transformative, challenging theoretical models. This dual approach represents the first decisive test capable of resolving the mechanisms of extreme stellar explosions, with profound implications for stellar evolution, early Universe dust budgets, interpretations of the LIGO black hole mass function, and future high-redshift survey strategies.

OBSERVING DESCRIPTION

AIMS: This proposal will obtain (1) deep imaging in 8 filters from F560W to F2100W to measure dust masses, and (2) medium-resolution spectroscopy targeting the [Ne II] 12.8 μm line to test for magnetar signatures in four of the most compelling (P)PISN candidates with peak luminosities in the regime of superluminous supernovae.

OBSERVING SETUP:

1) Imaging

-Sub-array: Full

-Dither pattern: 4-point cycling + large dither pattern size

-Read-out pattern: FASTR1

-Filters: F560W, F770W, F1000W, F1130W, F1280W, F1500W, F1800W, F2100W

-In each filter, the observation consists of a combination of Groups/Int, Integrations/Exp and Exposures/Dith that will allow us to reach a S/N of 20-40.

2) Spectroscopy

-Each observation is divided into an on-scene and off-scene epic.

-On-scene:

JWST Proposal 11027 (Created: Thursday, May 28, 2026, 7:00:22PM Eastern Standard Time) - Overview

- Dither pattern: 4-point cycling point-source pattern for 2018ibb and 2020abjc and extended-source pattern for 2021yjf and 2023iop.
 - Read-out pattern: FASTR1
 - Critical sub-bands: B for SNe 2018ibb, 2021yjf and 2023iop and C for SN 2020abjc (The targets have redshifts between 0.1 and 0.22. Depending on the redshift, the sub-band was chosen to cover [Ne II] 12.8 μm .)
 - Each observation consists of a combination of Groups/Int, Integrations/Exp and Exposures/Dith that will allow us to reach a S/N of 2-3 per resolution element (4xspectral binning for all objects).
 - SN2021yjf and SN2023iop: Simultaneous imaging will be performed in F770W to improve the astrometry.
- Off-scene:
- Dither pattern: 2-point cycling
 - Read-out pattern: FASTR1
 - Critical sub-bands: B for SNe 2018ibb, 2021yjf and 2023iop and C for SN 2020abjc
 - The background mode is used for SNe 2018ibb and 2020abjc. Owing to the higher source density in the fields of SNe 2021yjf and 2023iop, an offset by 7-25" in X and Y (each) from the target coordinates is applied to point at a blank field.
 - Simultaneous imaging will be performed in F770W (SN2021yjf and SN2023iop) or F1500W (SN2018ibb). No simultaneous imaging for SN2020abjc to maximise visibility.
 - Same number of groups like on-scene observation. Requested integration time is about 25% of the on-scene time, ensuring a good background spectrum.
- The imaging and spectroscopy observation of a given target are linked to each other to minimise slew overheads. Order of the observation: (1) MRS off-scene, (2) MRS on-scene, and (3) imaging.
- Timing of the observation: The observations can be done at any time during Cycle 5.
- Target Acquisitions: The target coordinates have an accuracy of <0.3 arcsec. JWST pointing accuracy is 0.1 arcsec. Hence, the science target will be close to the centre of the MRS FoV. No fine-tuning is requested during the target-acquisition.

Proposal 11027 - Targets - The last gasps of cosmic titans

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	SN2018ibb	RA: 04 38 56.9500 (69.7372917d) Dec: -20 39 44.10 (-20.66225d) Equinox: J2000 <i>Comments:</i> <i>Category=Star</i> <i>Description=[Supernovae]</i> <i>Extended=NO</i>	Epoch of Position: 2000	
(2)	SN2020abjc	RA: 09 28 0.2740 (142.0011417d) Dec: +14 07 16.62 (14.12128d) Equinox: J2000 <i>Comments:</i> <i>Category=Star</i> <i>Description=[Supernovae]</i> <i>Extended=NO</i>	Epoch of Position: 2000	
(3)	SN2021yfl	RA: 01 37 46.1710 (24.4423792d) Dec: -01 15 17.78 (-1.25494d) Equinox: J2000 <i>Comments:</i> <i>Category=Star</i> <i>Description=[Supernovae]</i> <i>Extended=NO</i>	Epoch of Position: 2000	
(4)	SN2023iop	RA: 00 14 42.6990 (3.6779125d) Dec: +21 51 33.77 (21.85938d) Equinox: J2000 <i>Comments:</i> <i>Category=Star</i> <i>Description=[Supernovae]</i> <i>Extended=NO</i>	Epoch of Position: 2000	
(5)	SN2018ibb-back	RA: 04 38 56.9500 (69.7372917d) Dec: -20 39 44.10 (-20.66225d) Equinox: J2000 <i>Comments:</i> <i>Category=Star</i> <i>Description=[Supernovae]</i> <i>Extended=NO</i>	Epoch of Position: 2000	
(6)	SN2018ibb-ima	RA: 04 38 56.9500 (69.7372917d) Dec: -20 39 44.10 (-20.66225d) Equinox: J2000 <i>Comments:</i> <i>Category=Star</i> <i>Description=[Supernovae]</i> <i>Extended=NO</i>	Epoch of Position: 2000	
(7)	SN2020abjc-back	RA: 09 28 0.2740 (142.0011417d) Dec: +14 07 16.62 (14.12128d) Equinox: J2000 <i>Comments:</i> <i>Category=Star</i> <i>Description=[Supernovae]</i> <i>Extended=NO</i>	Epoch of Position: 2000	

Proposal 11027 - Targets - The last gasps of cosmic titans

(8)	SN2020abjc-ima	RA: 09 28 0.2740 (142.0011417d) Dec: +14 07 16.62 (14.12128d) Equinox: J2000	Epoch of Position: 2000
<i>Comments:</i> <i>Category=Star</i> <i>Description=[Supernovae]</i> <i>Extended=NO</i>			
(9)	SN2023iop-back	RA: 00 14 42.6990 (3.6779125d) Dec: +21 51 33.77 (21.85938d) Equinox: J2000	Epoch of Position: 2000
<i>Comments:</i> <i>Category=Star</i> <i>Description=[Supernovae]</i> <i>Extended=NO</i>			
(10)	SN2023iop-ima	RA: 00 14 42.6990 (3.6779125d) Dec: +21 51 33.77 (21.85938d) Equinox: J2000	Epoch of Position: 2000
<i>Comments:</i> <i>Category=Star</i> <i>Description=[Supernovae]</i> <i>Extended=NO</i>			
(11)	SN2021yfi-back	RA: 01 37 46.5210 (24.4438375d) Dec: -01 14 50.78 (-1.24744d) Equinox: J2000	Epoch of Position: 2000
<i>Comments:</i> <i>Category=Star</i> <i>Description=[Supernovae]</i> <i>Extended=NO</i>			
(12)	SN2021yfi-ima	RA: 01 37 46.1710 (24.4423792d) Dec: -01 15 17.78 (-1.25494d) Equinox: J2000	Epoch of Position: 2000
<i>Comments:</i> <i>Category=Star</i> <i>Description=[Supernovae]</i> <i>Extended=NO</i>			

Proposal 11027 - Observation 1 - The last gasps of cosmic titans

Observation	Proposal 11027, Observation 1: SN2018ibb - MIRI - MRS - off-scene												Fri May 29 00:00:22 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
	Background Observation For: [SN2018ibb - MIRI - MRS - on-scene (Obs 2)]												
Diagnostics	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous		
	(5)	SN2018ibb-back	RA: 04 38 56.9500 (69.7372917d) Dec: -20 39 44.10 (-20.66225d) Equinox: J2000				Epoch of Position: 2000						
	Comments: Category=Star Description=[Supernovae] Extended=NO												
Acquisition	#	Target											
	1	NONE											
Template	AcqFilter	Primary Channel				Simultaneous Imaging			Imager Subarray		Grating Wheel Direction		
	F1500W	Imager				YES			FULL		Allow Auto Reorder		
Dithers	#	Dither Type				Optimized For				Direction			
	1	2-Point				BACKGROUND				NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1		IMAGER	F1500W	FASTR1	30	8	1	Dither 1	2	16	1370.87	
	1	MEDIUM(B)	MRSLONG		FASTR1	30	8	1	Dither 1	2	16	1370.87	271397.39
	1	MEDIUM(B)	MRSSHORT		FASTR1	30	8	1	Dither 1	2	16	1370.87	271397.39

Proposal 11027 - Observation 1 - The last gasps of cosmic titans

Special Requirements	Sequence Observations 1, 2, 3, Non-interruptible
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Proposal 11027 - Observation 2 - The last gasps of cosmic titans

Observation	Proposal 11027, Observation 2: SN2018ibb - MIRI - MRS - on-scene												Fri May 29 00:00:22 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
	Background Observations:[SN2018ibb - MIRI - MRS - off-scene (Obs 1)]												
Diagnostics	(SN2018ibb - MIRI - MRS - on-scene (Obs 2)) Warning (Form): The science and background exposures have inconsistent Simultaneous Imaging selections which may result in non-optimal science output.												
	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates					Targ. Coord. Corrections			Miscellaneous		
	(1)	SN2018ibb	RA: 04 38 56.9500 (69.7372917d) Dec: -20 39 44.10 (-20.66225d) Equinox: J2000					Epoch of Position: 2000					
	Comments: Category=Star Description=[Supernovae] Extended=NO												
Acquisition	#	Target											
	1	NONE											
Template	AcqFilter	Primary Channel				Simultaneous Imaging			Imager Subarray		Grating Wheel Direction		
	F1500W	All MRS				NO			FULL		Allow Auto Reorder		
Dithers	#	Dither Type					Optimized For			Direction			
	1	4-Point					POINT SOURCE			NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dither	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	MEDIUM(B)	MRSLONG		FASTR1	30	16	1	Dither 1	4	64	5494.579	271397.39
	1	MEDIUM(B)	MRSSHORT		FASTR1	30	16	1	Dither 1	4	64	5494.579	271397.39

Proposal 11027 - Observation 2 - The last gasps of cosmic titans

Special Requirements	Sequence Observations 1, 2, 3, Non-interruptible
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Proposal 11027 - Observation 3 - The last gasps of cosmic titans

Observation	Proposal 11027, Observation 3: SN2018ibb - MIRI - Imaging										Fri May 29 00:00:22 GMT 2026
	Diagnostic Status: Warning										
	Observing Template: MIRI Imaging										
Diagnostics	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(6)	SN2018ibb-ima	RA: 04 38 56.9500 (69.7372917d) Dec: -20 39 44.10 (-20.66225d) Equinox: J2000			Epoch of Position: 2000					
	Comments: Category=Star Description=[Supernovae] Extended=NO										
Template	Subarray										
	FULL										
Dithers	#	Dither Type	Starting Point	Number of Points	Points	Starting Set	Number of Sets	Optimized For	Direction	Pattern Size	
	1	CYCLING	1	4						LARGE	
Spectral Elements	#	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	F560W	FASTR1	20	3	1	Dither 1	4	12	688.21	271397.1
	2	F770W	FASTR1	20	3	1	Dither 1	4	12	688.21	271397.2
	3	F1000W	FASTR1	14	1	1	Dither 1	4	4	155.402	271397.3
	4	F1130W	FASTR1	14	1	1	Dither 1	4	4	155.402	271397.4
	5	F1280W	FASTR1	15	3	1	Dither 1	4	12	521.708	271397.5
	6	F1500W	FASTR1	9	6	1	Dither 1	4	24	654.909	271397.6
	7	F1800W	FASTR1	16	8	1	Dither 1	4	32	1498.522	271397.7
	8	F2100W	FASTR1	15	8	1	Dither 1	4	32	1409.72	271397.8
Special Requirements	Sequence Observations 1, 2, 3, Non-interruptible										

Proposal 11027 - Observation 4 - The last gasps of cosmic titans

Observation	Proposal 11027, Observation 4: SN2020abjc - MIRI - MRS - off-science												Fri May 29 00:00:22 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
	Background Observation For: [SN2020abjc - MIRI - MRS - on-science (Obs 5)]												
Diagnostics	(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates					Targ. Coord. Corrections			Miscellaneous		
	(7)	SN2020abjc-back	RA: 09 28 0.2740 (142.0011417d) Dec: +14 07 16.62 (14.12128d) Equinox: J2000					Epoch of Position: 2000					
	Comments: Category=Star Description=[Supernovae] Extended=NO												
Acquisition	#	Target											
	1	NONE											
Template	AcqFilter	Primary Channel				Simultaneous Imaging			Imager Subarray		Grating Wheel Direction		
	F1500W	All MRS				YES			FULL		Allow Auto Reorder		
Dithers	#	Dither Type					Optimized For			Direction			
	1	2-Point					BACKGROUND			NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1		IMAGER	F770W	FASTR1	30	8	1	Dither 1	2	16	1370.87	
	1	LONG(C)	MRSLONG		FASTR1	30	8	1	Dither 1	2	16	1370.87	271397.36
	1	LONG(C)	MRSSHORT		FASTR1	30	8	1	Dither 1	2	16	1370.87	271397.36

Proposal 11027 - Observation 4 - The last gasps of cosmic titans

Special Requirements	Sequence Observations 4, 5, 6, Non-interruptible
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Proposal 11027 - Observation 5 - The last gasps of cosmic titans

Observation	Proposal 11027, Observation 5: SN2020abjc - MIRI - MRS - on-science												Fri May 29 00:00:22 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
	Background Observations:[SN2020abjc - MIRI - MRS - off-science (Obs 4)]												
Diagnostics	(SN2020abjc - MIRI - MRS - on-science (Obs 5)) Warning (Form): The science and background exposures have inconsistent Simultaneous Imaging selections which may result in non-optimal science output.												
	(Visit 5:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates					Targ. Coord. Corrections			Miscellaneous		
	(2)	SN2020abjc	RA: 09 28 0.2740 (142.0011417d) Dec: +14 07 16.62 (14.12128d) Equinox: J2000					Epoch of Position: 2000					
	Comments: Category=Star Description=[Supernovae] Extended=NO												
Acquisition	#	Target											
	1	NONE											
Template	AcqFilter	Primary Channel				Simultaneous Imaging			Imager Subarray		Grating Wheel Direction		
	F1500W	All MRS				NO			FULL		Allow Auto Reorder		
Dithers	#	Dither Type					Optimized For			Direction			
	1	4-Point					POINT SOURCE			NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/E xp	Exposures/Dit h	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	LONG(C)	MRSLONG		FASTR1	30	16	1	Dither 1	4	64	5494.579	271397.36
	1	LONG(C)	MRSSHORT		FASTR1	30	16	1	Dither 1	4	64	5494.579	271397.36

Proposal 11027 - Observation 5 - The last gasps of cosmic titans

Special Requirements	Sequence Observations 4, 5, 6, Non-interruptible
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Proposal 11027 - Observation 6 - The last gasps of cosmic titans

Observation	Proposal 11027, Observation 6: SN2020abjc- MIRI - Imaging										Fri May 29 00:00:22 GMT 2026
	Diagnostic Status: Warning										
	Observing Template: MIRI Imaging										
Diagnostics	(Visit 6:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(8)	SN2020abjc-ima	RA: 09 28 0.2740 (142.0011417d) Dec: +14 07 16.62 (14.12128d) Equinox: J2000			Epoch of Position: 2000					
	Comments: Category=Star Description=[Supernovae] Extended=NO										
Template	Subarray										
	FULL										
Dithers	#	Dither Type	Starting Point	Number of Points	Points	Starting Set	Number of Sets	Optimized For	Direction	Pattern Size	
	1	CYCLING	1	4						LARGE	
Spectral Elements	#	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	F560W	FASTR1	20	8	1	Dither 1	4	32	1853.727	271397.9
	2	F770W	FASTR1	20	8	1	Dither 1	4	32	1853.727	271397.10
	3	F1000W	FASTR1	19	10	1	Dither 1	4	40	2208.932	271397.11
	4	F1130W	FASTR1	11	6	1	Dither 1	4	24	788.111	271397.12
	5	F1280W	FASTR1	19	10	1	Dither 1	4	40	2208.932	271397.13
	6	F1500W	FASTR1	17	10	1	Dither 1	4	40	1986.929	271397.14
	7	F1800W	FASTR1	39	18	1	Dither 1	4	72	7981.015	271397.15
	8	F2100W	FASTR1	23	26	1	Dither 1	4	104	6915.4	271397.16
Special Requirements	Sequence Observations 4, 5, 6, Non-interruptible										

Proposal 11027 - Observation 7 - The last gasps of cosmic titans

Observation	Proposal 11027, Observation 7: SN2021yfj - MIRI - MRS - off-scene												Fri May 29 00:00:22 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
	Background Observation For: [SN2021yfj - MIRI - MRS - on-science (Obs 8)]												
Diagnostics	(Visit 7:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous		
	(11)	SN2021yfj-back	RA: 01 37 46.5210 (24.4438375d) Dec: -01 14 50.78 (-1.24744d) Equinox: J2000				Epoch of Position: 2000						
	Comments: Category=Star Description=[Supernovae] Extended=NO												
Acquisition	#	Target											
	1	NONE											
Template	AcqFilter	Primary Channel			Simultaneous Imaging			Imager Subarray			Grating Wheel Direction		
	F1500W	All MRS			YES			FULL			Allow Auto Reorder		
Dithers	#	Dither Type				Optimized For				Direction			
	1	2-Point				BACKGROUND				NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1		IMAGER	F770W	FASTR1	20	5	1	Dither 1	2	10	577.208	
	1	MEDIUM(B)	MRSLONG		FASTR1	20	5	1	Dither 1	2	10	577.208	271397.38
	1	MEDIUM(B)	MRSSHORT		FASTR1	20	5	1	Dither 1	2	10	577.208	271397.38

Proposal 11027 - Observation 7 - The last gasps of cosmic titans

Special Requirements	Sequence Observations 7, 8, 9, Non-interruptible
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Proposal 11027 - Observation 8 - The last gasps of cosmic titans

Observation	Proposal 11027, Observation 8: SN2021yfj - MIRI - MRS - on-scene												Fri May 29 00:00:22 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
	Background Observations:[SN2021yfj - MIRI - MRS - off-scene (Obs 7)]												
Diagnostics	(Visit 8:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates					Targ. Coord. Corrections			Miscellaneous		
	(3)	SN2021yfj	RA: 01 37 46.1710 (24.4423792d) Dec: -01 15 17.78 (-1.25494d) Equinox: J2000					Epoch of Position: 2000					
	Comments: Category=Star Description=[Supernovae] Extended=NO												
Acquisition	#	Target											
	1	NONE											
Template	AcqFilter	Primary Channel				Simultaneous Imaging			Imager Subarray		Grating Wheel Direction		
	F1500W	All MRS				YES			FULL		Allow Auto Reorder		
Dithers	#	Dither Type					Optimized For			Direction			
	1	4-Point					EXTENDED SOURCE			NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1		IMAGER	F770W	FASTR1	20	10	1	Dither 1	4	40	2319.933	
	1	MEDIUM(B)	MRSLONG		FASTR1	20	10	1	Dither 1	4	40	2319.933	271397.38
	1	MEDIUM(B)	MRSSHORT		FASTR1	20	10	1	Dither 1	4	40	2319.933	271397.38

Proposal 11027 - Observation 8 - The last gasps of cosmic titans

Special Requirements	Sequence Observations 7, 8, 9, Non-interruptible
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Proposal 11027 - Observation 9 - The last gasps of cosmic titans

Observation	Proposal 11027, Observation 9: SN2021yfj - MIRI - Imaging										Fri May 29 00:00:22 GMT 2026
	Diagnostic Status: Warning										
	Observing Template: MIRI Imaging										
Diagnostics	(Visit 9:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(12)	SN2021yfj-ima	RA: 01 37 46.1710 (24.4423792d) Dec: -01 15 17.78 (-1.25494d) Equinox: J2000			Epoch of Position: 2000					
	Comments: Category=Star Description=[Supernovae] Extended=NO										
Template	Subarray										
	FULL										
Dithers	#	Dither Type	Starting Point	Number of Points	Points	Starting Set	Number of Sets	Optimized For	Direction	Pattern Size	
	1	CYCLING	1	4						LARGE	
Spectral Elements	#	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	F560W	FASTR1	13	2	1	Dither 1	4	8	299.704	271397.17
	2	F770W	FASTR1	20	3	1	Dither 1	4	12	688.21	271397.18
	3	F1000W	FASTR1	14	1	1	Dither 1	4	4	155.402	271397.19
	4	F1130W	FASTR1	8	2	1	Dither 1	4	8	188.703	271397.20
	5	F1280W	FASTR1	10	2	1	Dither 1	4	8	233.103	271397.21
	6	F1500W	FASTR1	8	3	1	Dither 1	4	12	288.604	271397.22
	7	F1800W	FASTR1	16	3	1	Dither 1	4	12	555.008	271397.23
8	F2100W	FASTR1	14	4	1	Dither 1	4	16	654.909	271397.24	
Special Requirements	Sequence Observations 7, 8, 9, Non-interruptible										

Proposal 11027 - Observation 10 - The last gasps of cosmic titans

Observation	Proposal 11027, Observation 10: SN2023iop - MIRI - MRS - off-scence												Fri May 29 00:00:22 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
	Background Observation For: [SN2023iop - MIRI - MRS - on-scence (Obs 11)]												
Diagnostics	(Visit 10:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous		
	(9)	SN2023iop-back	RA: 00 14 42.6990 (3.6779125d) Dec: +21 51 33.77 (21.85938d) Equinox: J2000				Epoch of Position: 2000						
	Comments: Category=Star Description=[Supernovae] Extended=NO												
Acquisition	#	Target											
	1	NONE											
Template	AcqFilter	Primary Channel				Simultaneous Imaging			Imager Subarray		Grating Wheel Direction		
	F1500W	Imager				YES			FULL		Allow Auto Reorder		
Dithers	#	Dither Type				Optimized For				Direction			
	1	2-Point				BACKGROUND				NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1		IMAGER	F770W	FASTR1	20	5	1	Dither 1	2	10	577.208	
	1	MEDIUM(B)	MRSLONG		FASTR1	20	5	1	Dither 1	2	10	577.208	271397.39
	1	MEDIUM(B)	MRSSHORT		FASTR1	20	5	1	Dither 1	2	10	577.208	271397.39

Proposal 11027 - Observation 10 - The last gasps of cosmic titans

Special Requirements	Sequence Observations 10, 11, 12, Non-interruptible
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Proposal 11027 - Observation 11 - The last gasps of cosmic titans

Observation	Proposal 11027, Observation 11: SN2023iop - MIRI - MRS - on-science												Fri May 29 00:00:22 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
	Background Observations:[SN2023iop - MIRI - MRS - off-science (Obs 10)]												
Diagnostics	(Visit 11:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates					Targ. Coord. Corrections			Miscellaneous		
	(4)	SN2023iop	RA: 00 14 42.6990 (3.6779125d) Dec: +21 51 33.77 (21.85938d) Equinox: J2000					Epoch of Position: 2000					
	Comments: Category=Star Description=[Supernovae] Extended=NO												
Acquisition	#	Target											
	1	NONE											
Template	AcqFilter	Primary Channel				Simultaneous Imaging			Imager Subarray		Grating Wheel Direction		
	F1500W	All MRS				YES			FULL		Allow Auto Reorder		
Dithers	#	Dither Type					Optimized For			Direction			
	1	4-Point					EXTENDED SOURCE			NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1		IMAGER	F770W	FASTR1	20	10	1	Dither 1	4	40	2319.933	
	1	MEDIUM(B)	MRSLONG		FASTR1	20	10	1	Dither 1	4	40	2319.933	271397.39
	1	MEDIUM(B)	MRSSHORT		FASTR1	20	10	1	Dither 1	4	40	2319.933	271397.39

Proposal 11027 - Observation 11 - The last gasps of cosmic titans

Special Requirements	Sequence Observations 10, 11, 12, Non-interruptible
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Proposal 11027 - Observation 12 - The last gasps of cosmic titans

Observation	Proposal 11027, Observation 12: SN2023iop - MIRI - Imaging										Fri May 29 00:00:22 GMT 2026
	Diagnostic Status: Warning										
	Observing Template: MIRI Imaging										
Diagnostics	(Visit 12:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(10)	SN2023iop-ima	RA: 00 14 42.6990 (3.6779125d)			Epoch of Position: 2000					
			Dec: +21 51 33.77 (21.85938d)								
			Equinox: J2000								
Template	Comments: Category=Star Description=[Supernovae] Extended=NO										
	Subarray										
	FULL										
Dithers	#	Dither Type	Starting Point	Number of Points	Points	Starting Set	Number of Sets	Optimized For	Direction	Pattern Size	
	1	CYCLING	1	4						LARGE	
Spectral Elements	#	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	F560W	FASTR1	8	2	1	Dither 1	4	8	188.703	271397.25
	2	F770W	FASTR1	8	2	1	Dither 1	4	8	188.703	271397.26
	3	F1000W	FASTR1	8	2	1	Dither 1	4	8	188.703	271397.27
	4	F1130W	FASTR1	8	2	1	Dither 1	4	8	188.703	271397.28
	5	F1280W	FASTR1	8	3	1	Dither 1	4	12	288.604	271397.29
	6	F1500W	FASTR1	8	4	1	Dither 1	4	16	388.506	271397.30
	7	F1800W	FASTR1	8	4	1	Dither 1	4	16	388.506	271397.31
	8	F2100W	FASTR1	8	4	1	Dither 1	4	16	388.506	271397.32
Special Requirements	Sequence Observations 10, 11, 12, Non-interruptible										